

Week: 6
Experiment: 4

“Isolation of DNA from Different Sources like Human Cheek Cells, Green Peas, Papaya, and Mango Samples”

Total Marks (30 Marks):

1) Lab Performance (10 Marks):

2) Observation (16 Marks):

a) For Papaya DNA Extraction:

Reagent	Effects and Observations
Washing Liquid	It breaks the cell membrane and nuclear envelope which releases DNA
1 ml of 5% of NaCl solution	Helps DNA precipitate by neutralizing its negative charge, allowing it to clump together
Papaya sample	Source of papaya cell
Isopropanol	DNA gets out of solution and is visible as white thread/ cloudy strands.

b) For PEAS DNA Extraction:

Reagent	Effects and Observations
5% NaCl Solution	Neutralizes the negative charges on DNA and helps proteins breaking, making DNA easier to separate.
Washing liquid	Breaks down Cell membranes and nuclear membranes. Solution appears to be slightly foamy and translucent.
Cold ethanol	separate layer above the aqueous extract. DNA, which is insoluble in cold ethanol, begins to precipitate into this alcohol layer. A white cloudy interface forms between alcohol and extract. After waiting for 5 mins, we observed visible dna strands.

c) For Human DNA Extraction:

Reagent	Effects and Observations
NaCl Solution	Neutralizes the negative charge on DNA, Helps breakdown protein chains that bind dna. Cheek cells from your mouth are dislodged and mixed into the saltwater.
Washing liquid	The detergent breaks open (lyses) the cell and nuclear membranes by dissolving their lipid bilayers, We noticed white big threads type structure forming in the water.
Isopropyl Alcohol	DNA will start to form and youll see small threads like structure of white colour. Cold alcohol reduces DNA solubility, causing it to precipitate.
Food Color	The food coloring just helps you see the DNA layer more clearly.

d) For Mango DNA Extraction:

Reagent	Effects and Observations
Mango Pulp	Source of DNA cell
Nacl Solution	Helps breaking down cell membrane and nucleas membrane.
Washing Liquid	The detergent breaks open (lyses) the cell and nuclear membranes by dissolving their lipid bilayers, Makes the solution foamy and less opaque
Cold ethanol	Cold ethanol (less dense) forms a separate top layer above the mango extract. DNA, which is insoluble in ethanol, starts to move into the ethanol layer. We observed orange liquid below, clear ethanol above with a cloudy foam forming. That were dna strands.

3) Results (3 Marks):

Saliva DNA Extraction

- After gargling with salt water and adding detergent, the solution appeared foamy with white small balls like structures.
- Upon adding cold ethanol, a layer formed, white strands of DNA became visible in test tube.
- These strands gradually clumped together and could be seen floating in solutions.

Green Peas DNA Extraction

- Grinding and filtering produced a green liquid extract.
- After adding NaCl and detergent, the solution became slightly translucent and foamy.
- When ethanol was added, a cloudy white layer appeared at the ethanol boundary.
- After a few minutes, white, stringy DNA fibers formed.

Mango Pulp DNA Extraction

- Filtering the pulp produced an orange yellowish liquid solution.
- Addition of NaCl and detergent caused slight foaming, but the solution color was still orange.
- After adding cold ethanol, a separate clear layer formed on top of the orange extract.
- In the test tube, white cloudy strands of DNA became visible.
- The DNA appeared as sticky, mucus like thread structure.

Papaya DNA Extraction

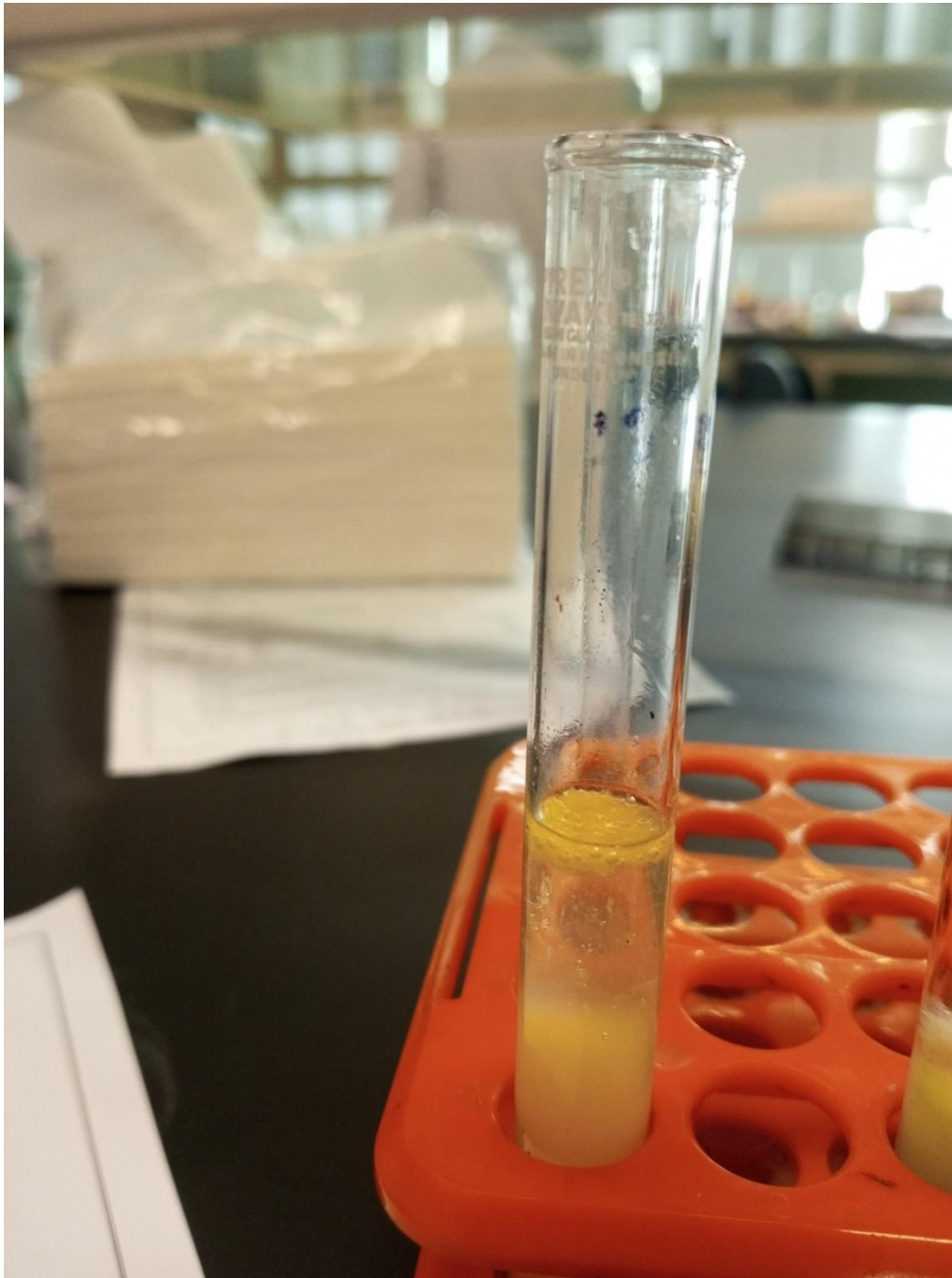
- Filtering produced a yellow liquid from the papaya pulp.
- After adding NaCl and detergent, the solution became slightly foamy while retaining its yellow color.
- When cold ethanol was added, two clear layers formed: the papaya extract at the bottom and ethanol at the top.
- Within a few minutes, a white, stringy, cloudy layer of DNA appeared.

4) Take pictures of the test tubes and paste them below (1 Mark):

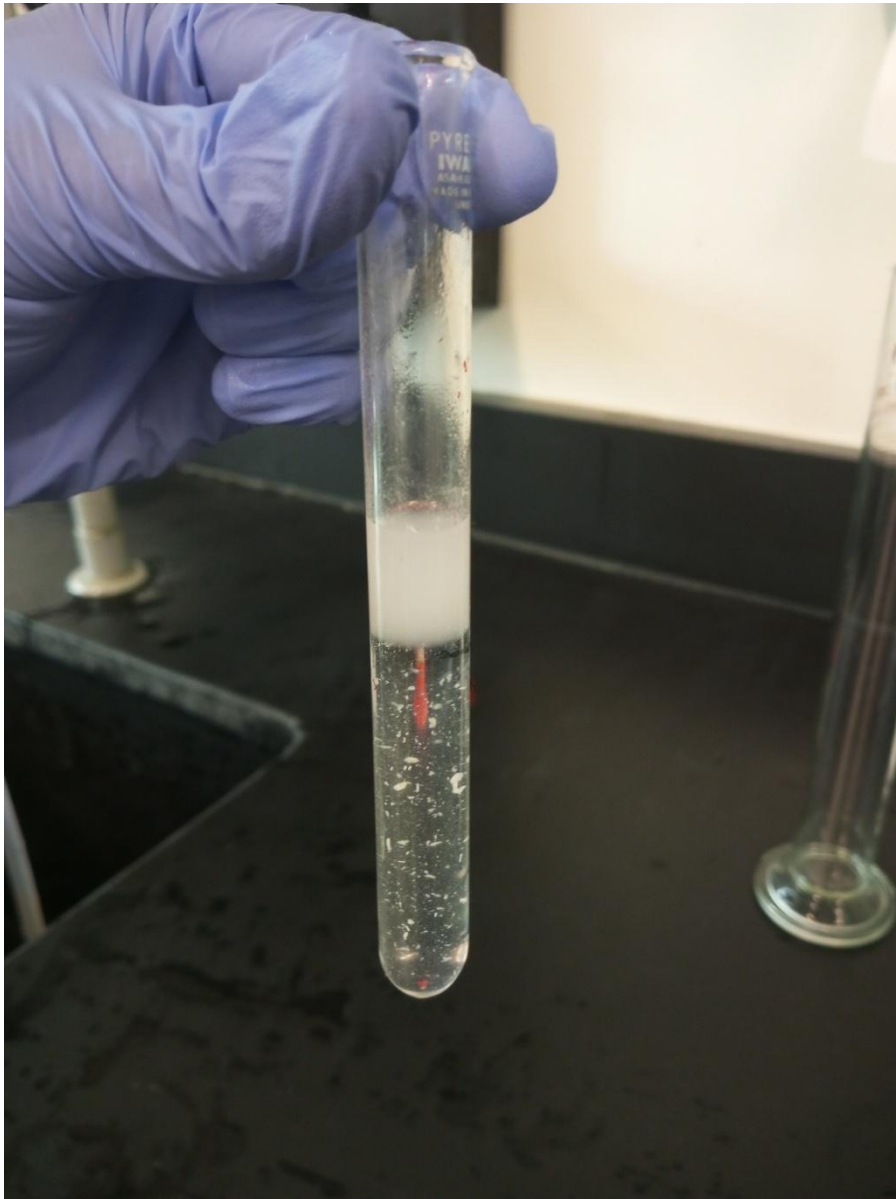
PEAS:-



MANGO:-



SALIVA:-



PAPAYA:-

